



LUNA 3

Our closest neighbor in space, the moon was an obvious target for early exploration. The first successful probe, the Russian-built Luna 3, flew past in October 1959 and sent back the first pictures of the far side of the moon, which permanently faces away from Earth. Throughout the 1960s, both the US and Russia sent a variety of orbiters and landers to investigate further.

Robotic probes

Human exploration of space has so far been limited to Earth orbit, but space agencies have sent dozens of automated space probes to explore planets, moons, comets, and asteroids. Flyby missions offer a brief glimpse of these objects, orbiters carry out more detailed surveys, and landers and rovers touch down on the surface to measure conditions and analyze rocks.

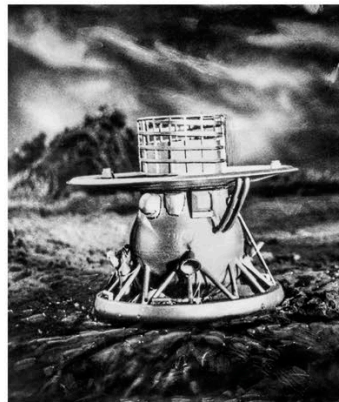


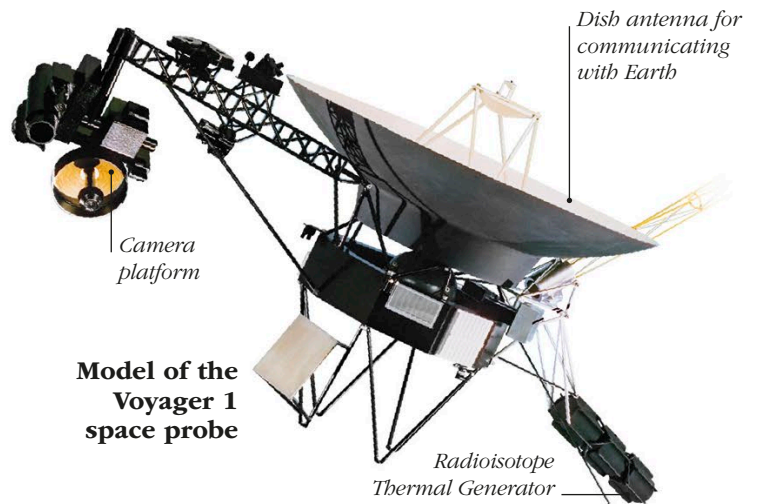
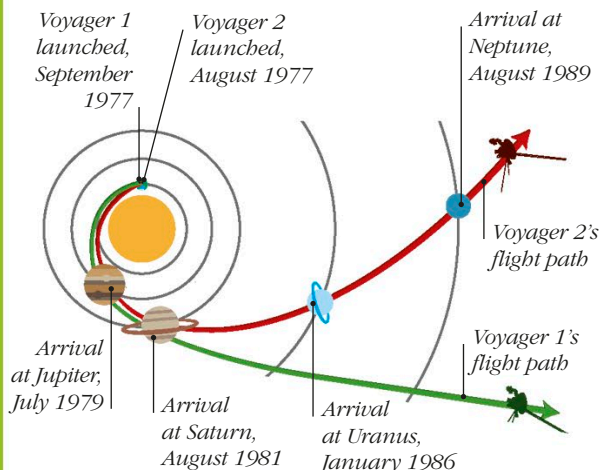
Illustration of
Venera 9 on Venus

THE VENERA MISSIONS

Venus is the nearest planet to Earth, but it proved a huge challenge to space probes. Several American and Russian probes made fleeting flybys or entered orbit, but early Russian landers were destroyed in Venus's hostile atmosphere. In 1975, Russia's heavily armored Venera 9 finally sent back the first picture of the volcanic landscape.

VOYAGER GRAVITY ASSIST

Even at high speeds, reaching the distant outer planets takes years. Fortunately, NASA engineers discovered a way of shortening journey times by using the gravity of other planets as a "slingshot" in a maneuver known as gravity assist. This enabled them to send two Voyager space probes on a tour of the outer solar system in the 1970s and 1980s.



Model of the
Voyager 1
space probe

POWER WITHOUT THE SUN

Probes in the inner solar system can use solar panels to make electricity, but missions venturing any further out from the sun than Mars need an alternative power supply. To solve the problem, NASA developed Radioisotope Thermal Generators (RTGs). These nuclear power supplies can produce electricity by harnessing the gentle heat emitted by small amounts of radioactive plutonium.